

RACHIT KUMAR TIWARI

Kolkata, West Bengal, India

CSE Undergraduate | Software & AI Developer

 GITHUB

MCKV Institute of Engineering, West Bengal

 HackerRank

 +91-9929243215

 tiwari.rachit@gmail.com

 rachit.unplugged@gmail.com

 LINKEDIN

 Portfolio

Technical Summary

B.Tech CSE student specializing in AI-integrated full-stack development. Built 10+ production-grade applications using Next.js, TypeScript, and Python — including a voice-first AI OS, a real-time privacy system with OpenCV, a clinical drug-interaction platform, and gesture-based HCI tools. Strong focus on scalable architecture, computer vision pipelines, graph algorithms, document processing, and LLM-powered workflows. Seeking software engineering roles to solve real-world problems through intelligent systems.

RESEARCH INTERESTS

Artificial Intelligence · Large Language Models (LLMs) · Computer Vision · Human-Computer Interaction · Real-Time Systems · Retrieval-Augmented Generation (RAG) · Privacy-Preserving Computing · AI for Social Good

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, TypeScript, C

Frameworks & Libraries: React, Next.js, Node.js, Electron, FastAPI, Tailwind CSS, shadcn/ui

Databases & Cloud: MongoDB, Firebase (Auth / Firestore), Supabase (Auth / Postgres / Storage)

Artificial Intelligence & Machine Learning: OpenAI API, Google Gemini API, Mistral AI, OpenCV, MediaPipe, Tesseract OCR, Face Recognition, RAG Pipelines

Developer Tools & Platforms: Git, GitHub, Docker, Postman, Figma, Vercel

Core Concepts: API Design, Real-Time Systems, WebSocket Communication, Computer Vision Pipelines, Graph Algorithms, LLM Orchestration, Document Processing, Containerization

PROJECT EXPERIENCE

•Vitals.io

August 2026

SaaS Unit Economics Analyzer for Startup Financial Intelligence

React, TypeScript, Node.js, Python, Pandas, D3.js

- Investigated automated financial data parsing pipelines capable of ingesting unstructured CSV datasets with fuzzy header auto-detection and schema inference.
- Developed an interactive analytics engine that transforms raw financial data into investor-grade metrics (LTV, CAC, payback period, and composite health scoring).
- Evaluated scenario modeling interfaces for startup unit economics, enabling dynamic parameter adjustment and real-time visualization of financial sustainability projections.

•MedCheck

August 2026

AI-Powered Clinical Drug Interaction & Safety Platform

React, Vite, FastAPI, Python, Mistral AI, Docker

- Built a clinical-grade drug interaction platform with a deterministic rule engine of 17 curated gold-standard pharmacology rules cross-referenced with OpenFDA labels, plus Mistral AI as an intelligent fallback for unstructured data.
- Engineered a FastAPI backend featuring multi-tier severity classification, a Stomach Guardian™ composite GI mucosal load score, Side Effect Radar with compounding risk detection, and JWT/bcrypt auth with guest sessions.
- Containerized with multi-stage Docker & Docker Compose; delivered a responsive React/Vite frontend with real-time search, color-coded severity indicators, and clipboard-export clinical summaries for provider visits.

•DocEasy

January 2026

Document Management & PDF Processing Platform

Next.js 16, TypeScript, Supabase, Tailwind CSS

- Built a professional-grade document manipulation platform with industrial PDF compression, intelligent browser/server routing for payloads up to 250MB, and O(1) memory complexity via Web Streams API.
- Engineered a hybrid processing architecture: lightweight tasks (image compression, basic PDF ops) run client-side via WebAssembly/Canvas; heavy workloads stream server-side with custom middleware bypassing the Next.js proxy.

- Implemented a document analysis engine featuring Resume Parsing, ATS scoring, and PDF summarization, backed by Supabase Auth (OAuth), Row-Level Security PostgreSQL, and secure cloud storage.

•WELCOMETOMYCITY

December 2025 - Present

Civic-Tech Travel & Itinerary Platform with Emotional Vibe Mapping

Next.js 15, TypeScript, MapLibre, Supabase

- Architected a Map-First Canvas travel platform using MapLibre GL JS with custom tilesets, dynamic GeoJSON routing, and glowing animated path layers for cinematic itinerary visualization.
- Built an Experience Engine scoring locations across emotional vectors (Romantic, Peaceful, Hidden Gem, Cultural) to dynamically reorder and highlight spots matching user mood.
- Implemented Supabase-backed persistence with Next.js Server Actions for zero-bundle-size mutations, optimistic UI updates, and secure database access without exposing API keys.

•Kred.ai

April 2026

Real-Time AI News Verification & Fact-Checking Platform

Next.js 16, TypeScript, Gemini API, Serper API

- Developed an AI-powered fact-checking engine combating misinformation by verifying news claims against live web evidence retrieved via Serper API, generating citations from trusted sources.
- Architected a Retrieval-Augmented Generation (RAG) pipeline combining real-time web search with Google Gemini for contextual claim analysis, producing evidence-backed verdicts with confidence scoring.
- Delivered multilingual verification support (English, Hindi, etc.) through a responsive, accessible UI with dark-mode-ready dashboards and mobile-first design using Tailwind CSS.

•PrivSight

December 2025 - Present

Real-Time Desktop Privacy System via Facial Recognition

Electron, React, TypeScript, Python, OpenCV

- Engineered a desktop privacy guard that automatically blurs/obscures sensitive screen content when unauthorized faces are detected in the webcam frame, using real-time facial recognition.
- Built a high-performance computer vision pipeline in Python with OpenCV, achieving ~30 FPS face detection with sub-100ms latency via optimized WebSocket communication between Electron frontend and Python backend.
- Ensured complete data privacy through 100% on-device processing — zero cloud dependency, zero data exfiltration, making it suitable for healthcare, legal, and enterprise environments.

•Air-Mouse & Gesture Controller

November 2025

Touchless HCI System (Cursor, Media & Air-Draw Control)

Python, OpenCV, MediaPipe, PyAutoGUI

- Developed two complementary gesture-driven HCI applications: a real-time air mouse enabling touchless cursor movement, click, drag, and scroll via 21-point hand landmark detection; and a media controller mapping gestures (pinch, swipe, fist) to system playback commands for Spotify, VLC, and YouTube.
- Implemented Kalman filtering for jitter-free cursor smoothing and moving average filters for stable swipe detection, achieving responsive, low-latency control usable with standard webcams.
- Integrated optional voice command support and an air-drawing canvas mode, demonstrating a modular, extensible architecture for human-computer interaction research and accessibility tooling.

•Kolkata Metro Route Planner

December 2025

Civic-Tech Graph-Based Transit Optimization Web App

Next.js 15, TypeScript, Tailwind CSS

- Built a full-featured metro route planning application covering all 6 Kolkata Metro lines, implementing Dijkstra's shortest-path algorithm with weighted edges for travel time, dwell time, and interchange penalties.
- Engineered a peak-hour mode applying a time-multiplier to journey estimates and a least-interchanges routing option, exposing configurable trade-offs between speed and convenience.
- Delivered an interactive SVG metro map with clickable stations, responsive mobile-first layout, and dark/light theme support — deployed on Vercel at metro-kolkata.vercel.app.

•OptraFi

October 2025

AI-Powered Personal Finance Platform

React, TypeScript, Firebase, OpenAI API

- Engineered a full-stack personal finance platform with real-time income/expense tracking, automated spending categorization, and interactive data visualizations for financial health monitoring.
- Implemented AI-driven investment recommendation engine analyzing user spending patterns to generate personalized portfolio suggestions with risk-adjusted insights.
- Architected secure, real-time dashboards using Firebase Authentication and Firestore, ensuring sub-second data sync across devices with role-based access control.

- Vocalis OS

January 2025 - Present

Voice-First Agentic AI Operating System

Next.js 16, TypeScript, Firebase, OpenAI API

– Architected a voice-first AI OS enabling hands-free task execution via speech recognition (Web Speech API) and text-to-speech synthesis, reducing interaction friction by enabling fully conversational workflows.

– Built autonomous AI agent system capable of real-time web search, contextual reminders, and dynamic to-do management through modular, extensible agent pipelines.

– Designed a scalable component architecture supporting pluggable AI agents with responsive, accessible UI across desktop and mobile viewports.

- Vitamend

August 2024 - Present

AI-Driven Medicine Donation & Redistribution Platform

Next.js 14, TypeScript, MongoDB, Gemini Vision

– Developed a secure medicine donation platform tackling pharmaceutical waste by enabling verified redistribution of unused medicines to underserved communities through AI-powered quality checks.

– Built a Gemini 1.5 Flash Vision OCR pipeline extracting medicine name, batch number, expiry date, and manufacturer from label images with confidence scoring and automated validation, flagging expired or unidentified products before pharmacist approval.

– Implemented end-to-end traceability using QR-code generation per donation batch, linking donors, verifiers, and recipients in an auditable chain of custody.

EDUCATION

- MCKV Institute of Engineering, West Bengal

2024 – 2028 (Expected May 2028)

B.Tech in Computer Science & Engineering

Cumulative YGPA: 8.12/10.0

Semester 3: 8.80/10.0

Semester 1: 8.62/10.0

Semester 2: 7.66/10.0

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Operating Systems, Discrete Mathematics, Software Engineering, Web Technologies

- St. Aloysius Orphanage and Day School, Kolkata

2022

Class XII (ISC)
- St. Aloysius Orphanage and Day School, Kolkata

2020

Class X (ICSE)

CERTIFICATIONS

HackerRank: SQL (Advanced, Intermediate & Basic), JavaScript (Intermediate), Node.js (Intermediate), REST API (Intermediate), Problem Solving (Intermediate), Java (Basic), Python (Basic), CSS (Basic)

ACHIEVEMENTS & LEADERSHIP

- Winner — Innovation Ideas 2024-25 (MCKVIE)| Project: Vitamend

August 2024 - October 2025

– Secured 1st place for an AI-driven medicine donation platform using Gemini 1.5 Flash Vision OCR, pharmacist verification workflows, and cryptographic chain-of-custody tracking.
- Winner — MCKVIE Research Symposium| Project: Vitamend

April 2026

– Awarded 1st place for research-driven innovation in pharmaceutical waste reduction, featuring automated donor-to-clinic routing via priority queues and real-time inventory tracking.
- Finalist — Hacklabify V1.0 | Top 8 Teams| Project: Vitals.io

August 2026

– Advanced to finals in the Startup Innovation Track by developing a SaaS unit economics analyzer that converts raw CSV financial data into investor-grade metrics (LTV, CAC, payback period, health score) with fuzzy header auto-detection and interactive scenario simulation.
- Finalist — Cosmohack | Top 10 Teams| Project: Vitamend

December 2025

– Ranked in the top 10 for an AI-assisted social impact platform addressing medicine accessibility through intelligent surplus redistribution and QR-based traceability.
- Finalist — HACKVSIT | Top 20 Teams| Project: Vitamend

July 2026

– Selected among top 20 teams nationwide for healthcare innovation, recognized for the end-to-end medicine verification pipeline and role-based access architecture.